USO Cheatsheet

Laboratoare

1. Lab 1
2. Lab 2
3. Lab 3
    - tar
    - zip
    - find
4. Lab 4
5. Lab 5
    - ip
6. Lab 6
7. Lab 7
    - ps extra
    - dd
8. Lab 8
    - cut and awk
    - tr and sed
    - for and if
9. Lab 9
    - users
    - permission
10. Lab 10
    - encryption
    - base64
    - hashing
11. Lab 11

Commands

Tar & Zip

```
tar cvf         # Add to archive
tar xvf         # Extract
tar tf          # Show contents

tar czvf        # Add to archive - compressed
tar xzvf        # Extract        - compressed (optional)
tar ztf         # Show contents  - compressed (optional)
```

```
zip filename.zip files      # Add to archive
zip -u filename.zip files   # Update archive
unzip filename.zip          # Extract
zip -sf                     # Show contents
```

Ps

```
# Sort every process by memory, shows only user,uid,pid,%mem,%cpu,rss,cmd
ps -e -o user,uid,pid,%mem,%cpu,rss,cmd --sort=%mem
```

Random

```
shuf -i 0-255 -n 1

/dev/urandom

$RANDOM
```

Create files with zero or random

```
dd if=/dev/urandom of=random bs=1M count=100    # File with random characters

dd if=/dev/zero of=zeros bs=1M count=69         # File with zeros
```

Mount files

```
mkfs.ext3                                     # Make an ext3 partition
mount -t ext3 filename mount/location/here  # Mount filename to that location
```

Random strings

```
# 20 character long strings
echo $RANDOM | md5sum | head -c 20; echo;
tr -dc A-Za-z0-9 </dev/urandom | head -c 20 ; echo;
pwgen 20 1
```

Word counter

```
wc -l file      # Line counter
wc -w file      # Word counter
wc -c file      # Character/byte counter
```

System info

```
 # Cpu info
lscpu
cat /proc/meminfo

# Memory info
lsmem
cat /proc/cpuinfo
free

lshw        # Hardware info
lspci       # PCI devices info
lsusb       # USB info
```

Users & groups

```
 sudo adduser                            # Longer more complex creation of user

sudo useradd -m gigel                   # Create user with username

sudo useradd -m -d /home/gigel gigel    # Create user with spicific home directory

sudo useradd -p $(openssl passwd -1 password) username  # User with password

w               # Show who is logged on and what they are doing
who             # Show who is logged on
whoami          # Show the users name
id              # Shows id, uid, groups of current user, or a specific user
finger gigel    # Display information about a specific user (empty means all)
passwd gigel    # Change the password of a specific user (empty means current user)

su gigel        # Switch to gigel user

sudo userdel gigel   # Deletes gigel from the computer :(
```

Chmod and chown

```
#   Read    |   Write   |   Execute
#     4           2           1

#   -rwxrwxrwx

#   User    |Users group|   Everyone else
#   rwx         rwx             rwx

chmod +x / +r / +w      # Adds read, execute or write persision to everyone
chmod 666               # Adds read and write to everyone

chown gigel tema-pclp   # Switches the ownership of tema-pclp to gigel
chown user:group file 1 # Switches the ownership of tema-pclp to user from the group group
```

SSH

```
ssh student@<ip-address>            # Connects to <ip-address> as student
ssh student@<ip-address> -p 2222    # Connects to <ip-address> as student on port 2222
ssh student@<ip-address>:~ ls       # Execute ls in home of student

scp /etc/passwd student@<ip-address>:~ # Copy passwd to the home directory of student
scp student@<ip-address>:/etc/passwd ~ # Copy passwd from remote to home directory
```

Encryption

```
# AES CBC type encryption with 256 bit key
openssl aes-256-cbc -in file.txt -out encrypted_file.enc -pass pass:"uso rules"

# Same but to decrypt
openssl aes-256-cbc -d -in encrypted_file.enc -out decrypted_file.txt -pass pass:"uso rules"

# Hashing
md5sum
sha256sum

echo -n "apples" | md5sum
```

Bash

Awk

```
awk '{print}' /etc/passwd                   # Prints the whole file
awk -F":" '{print $1}' /etc/passwd          # Prints the users
awk -F":" '/regex/ {print $1}' /etc/passwd  # Prints the line that match that regex
awk '{print NR,$0}' /etc/passwd             # Prints every line with the line number
awk -F":" '{print $1, $NF}' /etc/passwd     # Prints the first and last column
```

If

```
 if condition1
then
...
elif codtion2
then
...
else
...
fi
```

For & while loops

```
 # For loops from 1 to 10
for i in 1 2 3 4 5 6 7 8 9 10; do ... done
for ((i = 1; i <= 10; i++)); do ... done
for i in $(seq 1 10); do ... done
for i in $(seq -f "%02g" 1 10); do ... done

for f in *; do ... done                          # For loops for every file
for arg in $@; do ... done                       # For every argument

# For every user in passwd
for user in $(awk -F ":" '{print $1}' /etc/passwd);
do
...
done

# For every line in countries.csv
cat countries.csv | while read line
do
...
done

# While loop
while condition
do
...
done
```

Split string

```
IFS=":"     # Your separator

# Get's every argument from /etc/passwd
cat /etc/passwd | while read user password uid gid gecos home shell
do
...
done
```